

exactly 80 mm: _____

FIRST SOLVE

New here? Do the walkthrough once at cubepath.com/5x5x5/walkthrough/

NOTATION
R U **F** **L** **D** **B** = turn that face clockwise, looking at it from outside the cube.
' = counter-clockwise, **2** = half turn. Lowercase **r** = that face plus the layer behind it, turning together.
y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS
 algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run at as one unit

1 WHITE CROSS



White center **DOWN**. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out.
 Can't see it? Build the four white edges on the **YELLOW** face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top, turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
RUR'U'



white sticker FRONT
L'ULU



white sticker UP run the first one once to knock it out, then look again
RUR'U'

3 MIDDLE EDGES

Top edge with **NO** yellow, turn U until its front color matches the center under it. The arrow shows which slot it goes to.

goes RIGHT

U **RUR'U'** **y** **L'ULU**



goes LEFT
U' **L'ULU** **y'** **RUR'U'**

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ **RUR'** sexy move
 ■ **RUR'U** Sune
 ■ **R'FRF'** sledgehammer
 gap = change grip

FIRST SOLVE

New here? Do the walkthrough once at cubepath.com/5.vercel.app/. This card is the memory jogger.

NOTATION

R U F L D B = turn that face clockwise, looking at it from outside the cube.
 ' = counter-clockwise. 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together.
 y = spin the whole cube left. Stay centered. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's color must match the center it touches. No algorithm – work it out.
 Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top. Turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
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3 MIDDLE EDGES

Top edge with NO yellow: Turn U until its front color matches the center under it. The arrow shows which slot it goes to.

goes RIGHT
U' R'UR' y L'ULU



goes LEFT
U' L'ULU y' RUR'U'

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

RUR'U' sexy move
 RUR'U Sune
 R'RF sledgehammer
 gap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left to right
FRUR'U'F'



Hook hook pointing front-right
fFRUR'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow; rest clockwise
RU'R'U RU2R'



Anti-Sune $\frac{1}{2}$ yellow; rest counter-clockwise
RU2R' U'RUR'



PI $\frac{2}{3}$ yellow; headlights left
fFRUR'U'f' FRUR'U'F'

OLL CORNERS - continued



Headlights $\frac{2}{3}$ yellow; headlights at the back
R2UR'U2 RD'R'U2R'



2x Headlights $\frac{2}{3}$ yellow; headlights both sides
RU'R'U RU'R' URU2R'



Chameleon $\frac{2}{3}$ yellow next to each other
fUR'U'r' FR'F'



Bowtie $\frac{2}{3}$ yellow diagonal
fR'FRUR'U' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn on yellow, also finishes the case. Machine-checked, there is the worst.

Top not all yellow and nothing matches? Run Sune and read again.

No yellow edge at the end? Run the first one, then read again.

■ **RU'R'U** sexy move ■ **RU'R'U** Sune ■ **R'FR'** sledgehammer gap = change grip

2-LOOK

OLL CROSS - yellow edges



Line bar left-to-right
fRUR'U'f'

Hook hook pointing front-right
fRUR'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow; rest clockwise
RUR'U' RU2R'

Anti-Sune $\frac{1}{2}$ yellow; rest counter-clockwise
RU2R' U'RUR'

Pi $\frac{3}{4}$ yellow; headlights left
fRUR'U'f' fRUR'U'f'

OLL CORNERS - continued



Headlights $\frac{3}{4}$ yellow; headlights at the back
RD'RU2R' RD'R'U2R'

2x Headlights $\frac{3}{4}$ yellow; headlights both sides
RUR'U' RUR'R' URU2R'

Chameleon $\frac{3}{4}$ 2 yellow next to each other
rUR'U'r' FRF'

Bowtie $\frac{3}{4}$ 2 yellow diagonal
F'rUR'U'r' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, there is the worst.
Top not all yellow and nothing matches? Run Sune and read again.

■ RUR'U' sexy move
 ■ RUR'U'
 ■ R'FRF' sledgehammer
 gap = change grip

ONE-LOOK PI1

ADJACENT CORNER SWAP

exactly one face shows headlights



• T1: headlights: pairs B-left F-left
RUR'U' R'F R2UR'UR' RUR'F'

Ja L solid: pairs B-right F-left R-front
x R2FRF' RU2 rUr U2

Jb L solid: pairs B-left F-right R-back
RUR'F' RUR'U' R'F R2UR'U'

Aa L headlights: pairs F-right B-front
x LZD2 L'UL D2 L'UL'

Ab L headlights: pairs B-right R-back
x' LZD2 LUL' D2 LU'L

F1 L solid
R'U'F' RUR'U' R'F R2UR'UR' RUR'UR

One face shows headlights: two corners of that face match. Hold as shown, turn, then turn the top to finish.

RUR'U' sec.moves. **RUR'U'** Sym.Sym. **R'FRF'** slideschamber. **rao** = change dir.

ADJACENT CORNER SWAP

continued



Ra L1: headlights: pairs F-right R-back
RU'RU' RU' RU' DR UR'DR' R'U2R'



Rb L1: headlights: pairs B-left R-front
R2F RUR'U' FR U2RU2UR



Ga L1: headlights: pairs F-right R-back
R2UR'UR'U' RU'R2 U'D R'UR'D'



Gb L1: headlights: pairs R-back R-front
RU'R UD' R2UR'UR'U' RU'R2D



Gc L1: headlights: pairs B-right R-front
R2UR'UR'U' RU'R2 UD' RU'R'D



Gd L1: headlights: pairs R-front R-back
RUR' UD' R2UR'UR'U' UR'UR2D

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ONE-LOOK PLI

ADJACENT CORNER SWAP

exactly one face shows headlights

• T L headlights: pairs R-left F-left
RUR'U' RF R2UR'U' RUR'F'

• Ja L solid: pairs B-right F-left R-front
x R2FRF' RU2 rUr U2

• Jb L solid: pairs B-left F-right R-back
RUR'F' RUR'U' RF R2UR'

• Aa L headlights: pairs F-right B-front
x L2D2 LU'L D2 LU'L

• Ab L headlights: pairs B-right R-back
x' L2D2 LUL' D2 LU'L

• F L solid
RU'F' RUR'U' RF R2UR'U' H

One face shows two headlights: two corners of that face match. Hold as shown, then turn the top to finish.

■ **RURV** acv move ■ **RURU** turn ■ **RF'RF'** sliderchange **can** = change grip

ADJACENT CORNER SWAP

continued

• Ra L headlights: pairs F-left
RUR'U' RUR DR'UR'D' RU'2R'

• Rb L headlights: pairs B-left
R2F RURU' FR U2RU'2R

• Ga L headlights: pairs F-right
R2UR'UR'U' RU'R2 U'D R'UR'D'

• Gb L headlights: pairs R-back
RUR' UD' R2UR'URU' RU'R2D

• Gc L headlights: pairs B-right
R2UR'URU' RU'R2 UD' RU'R'D

• Gd L headlights: pairs R-front
RUR' UD' R2UR'URU' UR'UR'D

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day — one is the mirror of the other.

NNXX – BIG CUBES AND THE MAP

BIG CUBES: 4x4 and 5x5
 Reduce: place the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 2nd LAYER ONLY – never widen it.

last 2 edges
 4x4 PLL parity R U R' F' R' F' R slice out, run it, slice back – both cubes
 4x4 OLL parity 2 edge up 2R2 Uw2 2R2 Uw2 2 edge pairs swapped - half the time

4x4 OLL parity : 1 edge pair flipped - half the time

Rw U2 x	Rw U2	Rw U2	U2	Lw U2	Rw' U2	Rw U2	Rw' U2	Rw' U2	Rw'
<i>(one edge difference, slice out, run it, slice back)</i>									
Rw U2 x	Rw U2	Rw U2	3Rw U2	Lw U2	Rw' U2	Rw U2	Rw' U2	Rw' U2	Rw'

Fix parity during the last layer, before OLL and PLL – both parity algorithms move corners.

NOTATION



Each letter turns that face clockwise, seen from outside the cube – so from your seat, D and B look counter-clockwise = counter-clockwise. z = half turn.

x = middle slice, turning the way L turns.
 y = z = the whole cube, turning the way R/U and F turn. Lowercase f l = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT – use it until you know Suno Twist a yellow corner four turns at a time, turning ONLY the top face between corners.

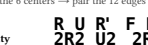
yellow faces RIGHT R'D'R'D x2
yellow faces FRONT R'D'R'D x2

WORDS

algorithm = a fixed sequence of turns / triggers – a short algorithm your hands run as one unit - sexy move = the R U R' U' trigger -
 sledgehammer = the R' F' R' trigger; parity = a case a 3x3 cannot have; edge pair = two edges glued into one, on a 4x4 or 5x5

alRUU' sexy move **RURU** Sune **R'FRF'** sledgehammer gap = change grips

ANNEX – BIG CUBES and THE MAP
BIG CUBES – 14x14 and 5x5
 Reduce: build the 6 centers – pair the 12 edges – solve it as a 3x3. Rr = 2 layers wide, slice back – both cubes
 2R = 2nd LAYER ONLY – never widen it.
 last 2 edges
 4x4 PLL parity



 slice out, 2 layers wide – both cubes
 2 edge pairs swapped – half the time
 4x4 OLL parity – 1 edge pair flipped – half the time
 Rw U2 x Rw U2 Rw U2 Rw U2 U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw' U2 Rw' U2 Rw' U2
 Rw U2 x Rw U2 Rw U2 3Rw' U2 Lw U2 Rw' U2 Rw U2 Rw' U2 Rw' U2 Rw' U2

NOTATION



U = up face, clockwise, 1/4 turn
 U' = up face, counter-clockwise, 1/4 turn
 U2 = up face, 1/2 turn
 M = middle slice, turning the way L turns
 x = y-z = the whole cube, turning the way R/U/D and U turn. Lowest case 2f = that face plus the layer behind it, turning together.

BEGINNER SHORTCUT – use it until you know Suno Twist a yellow corner four turns at a time, turning ONLY the top face between corners.

yellow faces RIGHT **R'D'R D' x2**
 yellow faces FRONT **D'R'D' x2**

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the SHORT edge. The number of copies must be exactly 80 mm: _____

